

**END TERM EXAMINATION**

FOURTH SEMESTER [B.TECH.] JULY-2023

Paper Code: EEC-206

Time: 3 Hours

Subject: Network Analysis and Synthesis

Maximum Marks: 75

Note: Attempt five questions including Q.No1 which is compulsory. Select one question from each unit.

Q.1 Attempt all parts:-

- (a) Differentiate between f-cut sets and f-tie sets.  
 (b) Synthesize the waveform as shown in figure 1.

(3x5=15)

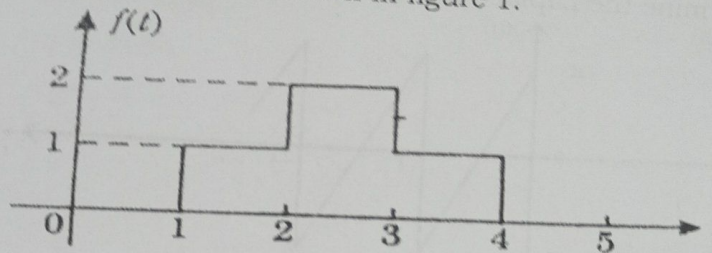


Figure: 1

- (c) Determine h-parameters in terms of Y-parameters?  
 (d) Check whether a polynomial expressed as  $P(s) = s^3 + 6s^2 + 11s + 6$  is Hurwitz or not.  
 (e) Differentiate between causal and non-causal system.

**UNIT-I**

- Q.2 a) For the graph shown in fig: 2, consider tree formed by branches (1, 2, 4), using this tree write  $A_c$ ,  $A$ ,  $B_f$  and  $Q_f$ . (15)

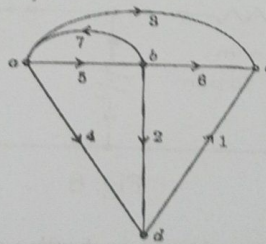


Fig: 2

- Q3 Find the equivalent Thevenin's equivalent circuit of the network shown in fig: 3. (15)

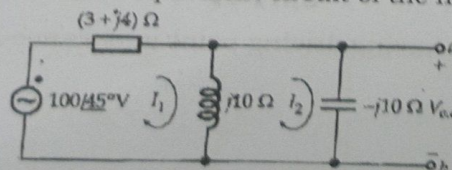


Fig: 3

P.T.O.



UNIT-II

- Q4 The circuit shown in fig: 4 is in the steady state with switch S closed. switch is opened at  $t=0$ . Determine  $i(t)$  in the circuit

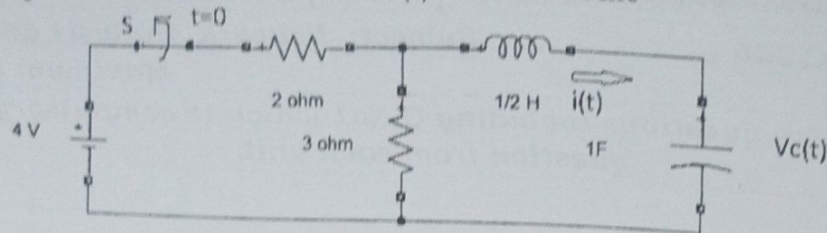


Fig: 4

- Q5 (a) Determine the Laplace transform of waveform shown in fig: 5. (7.5)

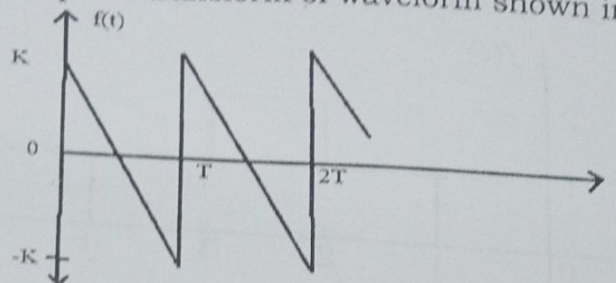


Fig: 5

- (b) Derive the expression for initial value and final value theorem. (7.5)

- Q.6 Obtain the Z-parameters of fig: 6? (15)

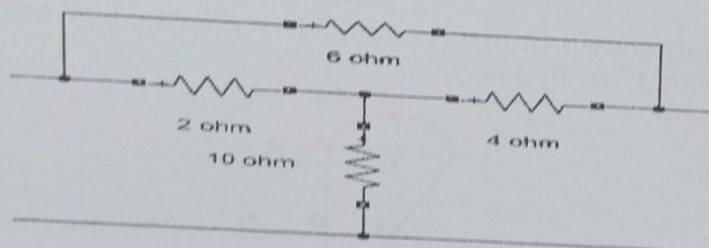


Fig: 6

- Q.7 a) Drive input and output impedance in terms of h-parameters.  
b) Briefly Drive the expression series-parallel connection. (7.5)

UNIT-IV

- Q. 8 Synthesize the network having driving point impedance  $Z(s) = \frac{2s^5 + 10s^3 + 10s}{s^4 + 4s^2 + 3}$  in the first Cauer form. (15)

- Q. 9 Write a short note on  
a) Band pass filter  
b) High pass filter (7.5x2=15)

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